

JEE MAIN REVISION MASTER: LAWS OF MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 152 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Laws Of Motion - Core Application	152 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Laws Of Motion - Core Application (Very High Yield — 152 PYQs)

Core Principle: Key problem pattern in Laws Of Motion.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionl..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "A small block of mass m slides down from the top of a frictionless inclined surface, while the inclined plane is moving towards left with co..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "A block is sliding down on an inclined plane of slope θ and at an instant $t=0$ this block is given an upward momentum so that it sta..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionless pulley which is fixed on the edge of a cube (S) , as shown in the figure below. Block P is positioned on the top surface which has no friction and block Q is in contact with side-surface, having coefficient friction μ . The cube (S) moves towards the right with acceleration of $(g / 2)$, where g is gravitational acceleration. During this movement the block P and Q remain stationary. The value of μ is ____ (take $g=10\text{m/s}^2$)

- (A) 0.33
- (B) 0.67 **[Correct]**
- (C) 1
- (D) 0.5

Correct Answer: B

Solution Analysis: Since the cube is accelerating at $a=(g / 2)$ to the right, when analysing the blocks from the cube's non-inertial frame, we must apply a pseudo-force ($F_{(p)}=m \times a$) on both blocks in the opposite direction to the acceleration of block S. For block P (Mass $m_{(P)}=2\text{kg}$) Along horizontal the tension (T) in the thread pulls it to the right. The pseudo-force ($m_{(P)} \cdot a$) pulls it to the left. Along vertical direction gravity ($m_{(P)}g$) acts downward and normal force ($N_{(P)}$) from the top surface acts upward. Since the block is stationary relative to the cube : $T=m_{(P)} \cdot a$